

Reviewer Pack — Minimal Rank of Formal Contexts vs AC0 Circuit Size

Entry 8c4ea325b38d · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Formal Contexts vs AC0 Circuit Size

Entry ID: 8c4ea325b38d **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-24 02:09:14 UTC

1. Conjecture

Field A (mathematical branch): Formal Concept Analysis (FCA) **Field B** (complexity object): Complexity Theory: AC0 Circuit Complexity

Statement:

[“For any boolean function f computable in AC0 complexity, the minimal rank of its associated formal context is upper bounded by a function of the size of f ’s input.”, ‘Equivalently, for all n -bit AC0 circuits C with depth d and gate count g , there exists a formal context with at most $O(n^2)$ concepts whose Galois connection can encode C .’, ‘Moreover, this bound holds for a random seed with probability greater than $3/4$.’]

Rationale (proposer’s reasoning):

[‘Formal concept analysis provides a framework to study binary relations that is closely related to information theory and combinatorial structures. The rank of the formal context associated with a function could potentially serve as an invariant that characterizes the complexity of the function.’, ‘If proven, this conjecture would provide a direct link between FCA and circuit complexity, offering a new perspective on understanding the complexity classes P and NP.’, ‘The use of formal contexts in complexity theory is rare, making this connection novel and potentially fruitful.’]

Taxonomy category: ACC_SIPSER (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): c1fa677a4ee55864

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The minimal rank of a formal context associated with an AC0 circuit is supported if it does not exceed $O(n^2)$ concepts for all $n \leq 40$ and at least 24 out of 30 random seeds, and falsified if any seed produces a minimal rank exceeding $O(n^2)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - formal concept analysis AND AC0 circuit complexity - AC0 circuit size AND formal context minimal rank - Galois connection encoding AND n-bit AC0 circuits IN formal context

Top relevant hits considered: - [<http://arxiv.org/abs/1210.2401v1>] Distributed Formal Concept Analysis Algorithms Based on an Iterative MapReduce Framework - [<http://arxiv.org/abs/1107.2822v1>] A Survey on how Description Logic Ontologies Benefit from Formal Concept Analysis - [<http://arxiv.org/abs/1809>] The Origins Space Telescope (OST) Mission Concept Study Interim Report - [<http://arxiv.org/abs/2304.02770v2>] Tight Correlation Bounds for Circuits Between AC0 and TC0 - [<http://arxiv.org/abs/0705.3525v1>] Comment on “Minimal size of a barchan dune” - [<http://arxiv.org/abs/2309.11491v3>] Vacuum Spacetime With Multipole Moments: The Minimal Size Conjecture, Black Hole Shadow, and Gravitational Wave Observab - [<http://arxiv.org/abs/1512.07347v1>] Galois Self-Dual Constacyclic Codes - [<http://arxiv.org/abs/2105.14538v1>] Longer Version for “Deep Context-Encoding Network for Retinal Image Captioning”

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import sys
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def formal_context(f, n):
        X = list(range(2**n))
        Y = list(range(len(f)))
        return X, Y

    def galois_connection(X, Y):
        L = []
        U = []
        for x in X:
```

```

        L.append([y for y in Y if all(x & (1 << i) == f(y)[i] for i in range(n))])
    for y in Y:
        U.append([x for x in X if all(x & (1 << i) == f(y)[i] for i in range(n))])
    return L, U

def minimal_rank(L):
    rank = 0
    while L:
        rank += 1
        new_L = []
        for l in L:
            if not any(l[i].issubset(new_l) for i, new_l in enumerate(new_L)):
                new_L.append(l)
        L = new_L
    return rank

n_values = [5, 10, 15, 20, 30, 40]
total_rank = 0
instances_tested = 0

for n in n_values:
    f = generate_boolean_function(n)
    X, Y = formal_context(f, n)
    L, U = galois_connection(X, Y)
    rank = minimal_rank(L)
    total_rank += rank
    instances_tested += 1

mean_rank = Fraction(total_rank, instances_tested)
conjecture_holds = mean_rank <= n**2
counterexample = "" if conjecture_holds else f"Mean rank {mean_rank} exceeds 0(n^2)"

return {
    "metric_name": "Minimal Rank",
    "metric_value": float(mean_rank),
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2**i - 1 for i in range(5, 31)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, **result}}")
        results.append(result)

    mean_rank = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_rank} std=0.0 support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])

```

```

    print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\\" first_failing_seed={first_failing_seed}")
else:
    print("RESULT: INCONCLUSIVE reason=unknown")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca65035d.py", line 78, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca65035d.py", line 56, in run_trial
    L, U = galois_connection(X, Y)
           ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca65035d.py", line 33, in galois_connection
    L.append([y for y in Y if all(x & (1 <= i) == f(y)[i] for i in range(n))])
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca65035d.py", line 33, in <genexpr>
    L.append([y for y in Y if all(x & (1 <= i) == f(y)[i] for i in range(n))])
              ^^^^^
TypeError: 'list' object is not callable

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from verifying whether the minimal rank of a formal context associated with an AC0 circuit is supported or falsified. | next: Re-run the test without errors to verify the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	10950
2	propose	ollama_remote	glm4:latest	0	0	9470
3	propose	ollama_remote	glm4:latest	0	0	9903
4	preregistration	ollama_remote	glm4:latest	0	0	5446
5	novelty	ollama_remote	glm4:latest	0	0	4562
6	novelty	ollama_remote	glm4:latest	0	0	6463
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13198

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9280
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10134
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8708
11	judge	ollama_remote	glm4:latest	0	0	23791

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 111904 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in research/audit/8c4ea325b38d.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/8c4ea325b38d.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/8c4ea325b38d.tar.gz (if generated)